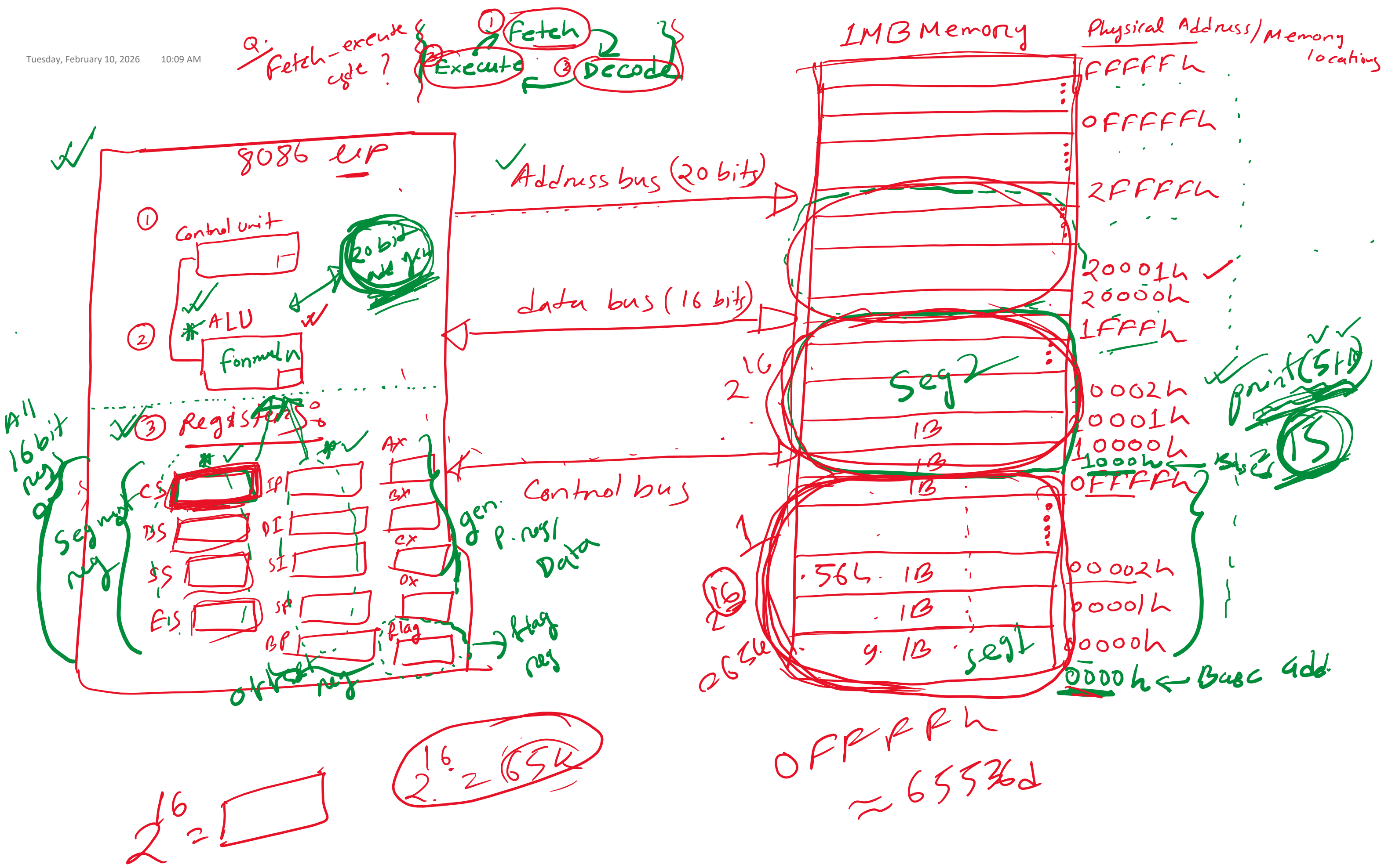

Q. Fetch-execute cycle?
① Fetch
② Decode
③ Execute
8086 uP
① Control unit
② ALU
formula
20 bit add. cal
③ Registers
All 16 bit reg
Segment reg
CS
DS
SS
ES
IP
DI
SI
SP
BP
AX
BX
CX
DX
flag
gen. p. reg / Data
flag reg
offset reg
Address bus (20 bits)
data bus (16 bits)
Control bus
1MB Memory
Physical Address / Memory locations
FFFFFh
0FFFFFh
2FFFFh
20001h
20000h
1FFFFh
10002h
10001h
10000h
1000h
0FFFFh
00002h
00001h
00000h
0000h ← Base add.
seg2
seg1
1B
2^{16}
65536
print (5th)
15
$2^{16} = 65K$
$2^{16} =$
0FFFFh
≈ 65536d

Segmentation: This is a process using which main memory of a computer is logically divided into different segments and each segment has its own base address.

④ 8086 uP has total 16 segments/partitions in its memory. ??

④ segmentation → 2 types:
① non-overlapping
② Overlapping

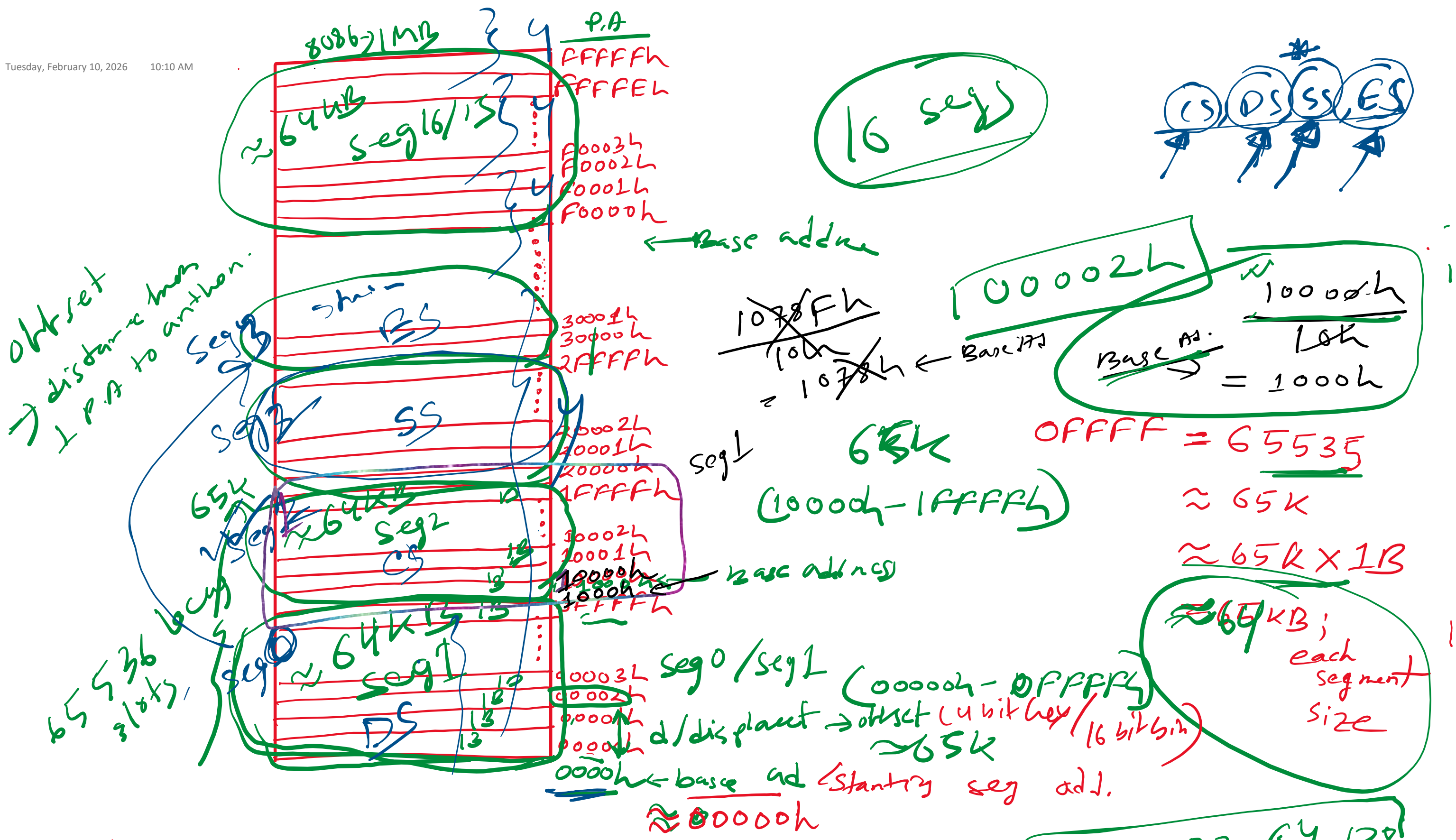
$$\text{Partition no./seg no.} = \frac{\text{main/RAM Memory size}}{\text{partition size}} = \frac{1\text{MB}}{2^{16}}$$

≈ size of the reg.

$$= \frac{1{,}048{,}576}{65536}$$

$$\approx 16$$

$2^{16} \simeq 65KB$


8086→1MB
P.A
FFFFFh
FFFFEh
F0003h
F0002h
F0001h
F0000h
← Base address
≈64KB
seg16/15
offset → distance from 1 P.A to another
ES
Seg3
SS
Seg2
65K
64KB
seg2
CS
Seg1
65536 locs/slots
seg0
≈64KB
seg1
DS
30001h
30000h
2FFFFh
20002h
20001h
20000h
1FFFFh
10002h
10001h
10000h
1000h
0FFFFh
00003h
00002h
00001h
00000h
0000h ← base ad / starting seg add.
≈00000h
d/displacement → offset (4 bit hex / 16 bit bin)
≈65K
seg0/seg1 (00000h - 0FFFFh)
seg1
65K
(10000h - 1FFFFh)
base address
16 segs
CS DS SS ES
~~1078Fh~~
~~10h~~
= ~~1078h~~ ← Base Ad
100002h
10000h
10h
Base Ad. →
= 1000h
0FFFF = 65535
≈ 65K
≈ 65K × 1B
≈64KB; each segment size

8, 16, 32, 64, 128

Base 2

Q: 2) ⊛ Segmentation is done to enhance the speed of execution of computing systems, so that up can fetch and execute data from memory fast and easily.

⊛ each segment has a size of 64KB for 8086 up

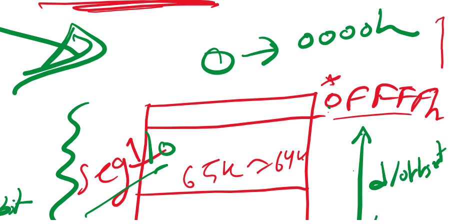
⊛ each segment is identified as a 16 bit number called segment number. Starting from 0000h – F000h. Segment register (CS, DS, SS, ES) holds them.

⊛ Within a segment, memory location/physical address is specified/identified with the help of an offset/displacement: offset registers → (IP, SI, DI, SP, BP). It is also a 16 bit number
hex

⊛ for 8086 up lowest offset is 0000h and highest offset is FFFFh.

⊛ logical address → segment : offset

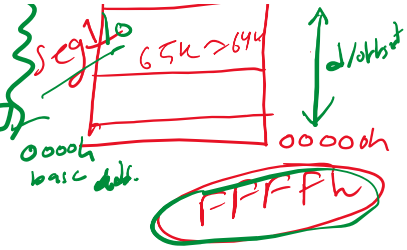
16 bit 16 bit binary . 16 bit binary 16 bit


0 → 0000h
0FFFFh
seg 0
65K→64K
d/offset

① logical address → segment : offset

16 bit binary number : 16 bit binary number

4 bit Hex number : 4 bit Hex number



② Physical address → 20 bit binary number 5 bit hex number

~~add~~ ALU

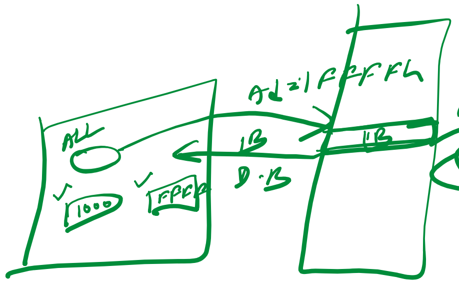
③ Physical address = segment * 10h + offset

$= 1000 * 10h + FFFFh$

$= 10000 + FFFFh$

$= 1FFFFh$

$= 1FFFFh$



mem.

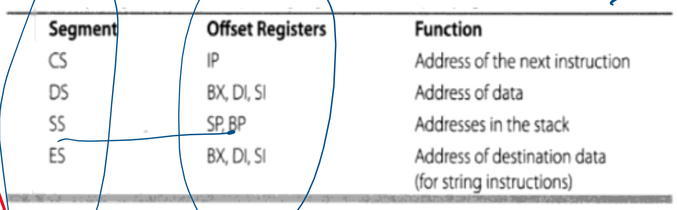
seg : offset

① CS : IP

② DS: DI/SI/BX

③ SS: SP/BP

④ ES: BX/DI/SI



Segment	Offset Registers	Function
CS	IP	Address of the next instruction
DS	BX, DI, SI	Address of data
SS	SP, BP	Addresses in the stack
ES	BX, DI, SI	Address of destination data (for string instructions)

Last but imp.

The starting physical address of a segment should be divisible by 10h.

1. Deduce the size of the address bus if the total size of the memory is 4 MB

2. Determine the total memory size if the address bus size is 21 bits.

1.

$4MB = 4 \times 1024 \times 1024$

$= 2^2 \times 2^{20}$

$= 2^{22}$

$\log_2(4000000) \approx 22$

$1KB = 2^{10} = 1024B$

$= 2^{22}$

$= \log_2(2^{22})$

$= 22$ bits Address bus

x) 1KB = 2^{10} = 1024B

$2^{10} \times 2^{10}$

2.

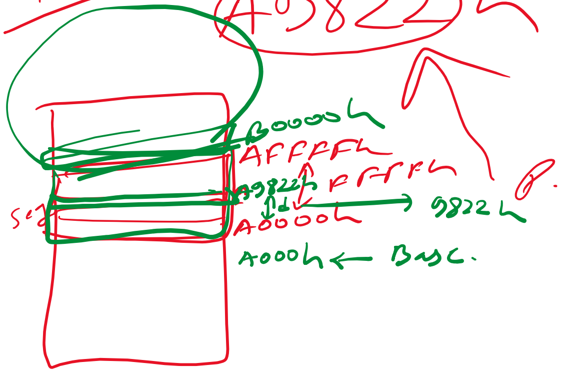
$2^{21} = 2 \times 2^{20} = 2$ MB

P2.

A9822h → seg : offset

log^x ~

A000h : 9822h


B0000h
AFFFFh
A9822h
FFFFh
9822h
seg
A0000h
A000h ← Base.

P.A = A000 * 10h + 9822h